

Rosen — §8.5: Inclusion–Exclusion
Selected Exercises

Alston

Semester 2, 2026

Exercise 13

How many bit strings of length eight do not contain six consecutive 0s?

Exercise 14

How many permutations of the 26 letters of the English alphabet do not contain any of the strings *fish*, *rat*, or *bird*?

Exercise 21

Write out the explicit formula given by the principle of inclusion–exclusion for the number of elements in the union of six sets when it is known that no three of these sets have a common intersection.

Exercise 22

Prove the principle of inclusion–exclusion using mathematical induction.

Exercise 29

Find a formula for the probability of the union of n events in a sample space.